

HYBRID DEEP LEARNING AND LARGE LANGUAGE MODEL PIPELINE FOR BIOMEDICAL IMAGE ANALYSIS.

GitHub Link: <https://github.com/Lojana96/Hybrid-Biomedical-Image-Analysis>

Student ID: 24106862

Student Name: Lojana Jegatheeswaran

1. Overview and Method

This project developed a complete pipeline for analyzing synthetic fluorescence microscopy images of cell nuclei. The goal was to compare three methods: direct image description using a vision-language model (VLM), classical image processing with a Numbers-First LLM, and U-Net deep learning segmentation. The final system integrates U-Net segmentation, deterministic feature extraction, and a local LLM to create a structured and auditable output.

The dataset consisted of 112 synthetic microscopy image-mask pairs, split into 80 for training, 20 for validation, and 12 unseen test images. All images were converted to grayscale, resized to 256 X 256 pixels, and normalized to the range [0,1]. The dataset included sparse, normal, dense, and clustered nucleus distributions.

=== Dataset Summary by Density Regime ===

	Density	Images	Mean nuclei count	Mean area coverage
0	Sparse	20	8.15 ± 1.76	2.25% ± 0.77%
1	Normal	56	26.27 ± 6.83	7.01% ± 1.99%
2	Dense	18	67.06 ± 13.56	16.94% ± 3.15%
3	Clustered	18	44.06 ± 8.65	9.19% ± 1.56%
4	Overall	112	32.45 ± 20.12	8.10% ± 4.86%

Table 1: Dataset Characteristics

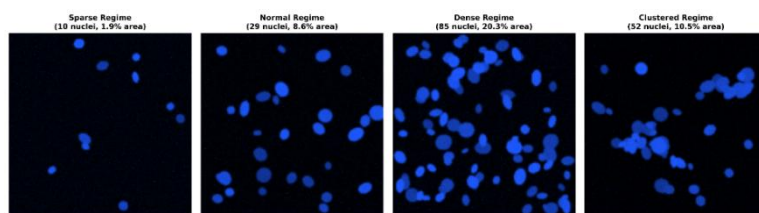


Figure 1: Representative fluorescence microscopy images showing sparse, normal, dense and clustered nucleus distributions.

The final hybrid architecture is:

Raw 256 X 256 image → U-Net → binary mask → connected-components-and-region-props → numerical-summary → local-LLM → Validated JSON and Narrative.

The U-Net used four encoder levels with 16, 32, 64, and 128 filters, 2 X 2 max pooling, transpose convolutions, and encoder-decoder skip connections. Classical analysis used Otsu thresholding, followed by morphological closing, hole filling, small-object removal, and connected-component analysis. For the illustrated train_000 image, the Otsu threshold was approximately .136.

2. VLM and Numbers – First Interpretation

Direct VLM

The first experiment sent a microscopy image directly to a multimodal model. Llama3.2-vision was requested, but the local Ollama backend could not load the mllama architecture. As a result, the fallback used iilava:7b, and this model information was recorded in the output for provenance.

The initial prompt was: **Naïve prompt:** "Look at this microscopy image and describe what you see." This gave a fluent response, but it also included unsupported biological interpretation. For instance, the model suggested that the cells undergo colony formation and could be fibroblasts. These statements were not supported by the synthetic image itself. This showed that a visually plausible VLM answer is not always reliable.

The prompt was then redesigned: **Optimized VLM prompt:** "You are a biomedical imaging analysis assistant. Your task is purely descriptive and objective. Do not provide clinical diagnosis, pathological staging. Disease classification, or speculative disease attribution. Analyse only information visually supported by the supplied microscopy image. Return valid JSON containing modality, tissue type, notable features, image quality, uncertainty, and summary narrative. Use uncertain whenever the visual evidence is insufficient."

At temperature 0.0, this produced a valid structured record identifying fluorescence microscopy, cellular nuclei, and features related to signal intensity, spatial distribution, and clustering. The

code also recorded `json_valid = True` and `schema_complete = True`.

Classical features and Numbers - First LLM

The second approach did not let the LLM view the original image. Instead, quantitative features were extracted first using Otsu segmentation and `skimage.measure.regionprops`.

For `train_000`, the pipeline detected **9 connected components**, with a total area of **1282 pixels**, mean area **142.4 ± 60.6 pixels**, mean eccentricity 0.504, mean solidity 0.956, mean foreground intensity 0.258, and field coverage 1.96%.

The numerical summary was then passed to `illama3.2`.

Optimized Numbers-First prompt: "you are a biomedical data-analysis assistant auditing quantitative measurements extracted from segmented microscopy images. You will not see the original image. Use only the numerical measurements supplied. Do not invent visual findings, diagnoses, diseases, tissue identities or measurements. Produce a factual narrative followed by JSON containing `n_objects`, `density_class`, `shape_regularity`, `quality_flag`, and `rationale`. If the evidence is insufficient, use uncertain."

The final parser was able to recover and validate the structured JSON even when the raw model response was incomplete. For this example, the final record contained `n_objects = 9`, `density_class = sparse`, `shape_regularity = regular`, and `quality_flag = pass`.

Q1. Which description is more useful, and which is more trustworthy?

The direct VLM is more useful for a quick general description, as it can describe the image without segmentation. However, the Numbers-First approach is more trustworthy because the key facts are calculated before the LLM is used.

Direct VLM produced unsupported biological statements. In contrast, the Numbers-First LLM received values such as area, count, eccentricity, and solidity that can be traced back to the segmentation output. For this reason, I would use the direct VLM only for supporting visual description and rely on the Numbers-First pipeline for quantitative reporting.

3.U-Net Training and Comparison with Otsu

The 4-level U-Net was trained for 15 epochs using Adam, learning rate .001, batch size 8, random-flip augmentation, and seed 42. Three loss functions were compared: Binary Cross-Entropy (BCE), Soft Dice and BCE + Dice.

Loss Ablation Summary Table (Computed from CSV):

	loss_function	final_train_loss	final_val_loss	final_val_dice	final_val_iou	peak_val_dice	peak_val_iou	peak_epoch
0	Combined (BCE + Dice)	0.7187	0.7820	0.9901	0.9804	0.9902	0.9807	13
1	BCE Only	0.1824	0.1861	0.9934	0.9869	0.9938	0.9877	13
2	Soft Dice Only	0.4780	0.6479	0.9673	0.9373	0.9738	0.9490	13

Table 2: U-Net loss- function ablation results computed from the training history.

BCE-only achieved the highest peak validation Dice (0.9938), followed by combined BCE + Dice (0.9902), while soft Dice achieved a lower peak score (0.9738)

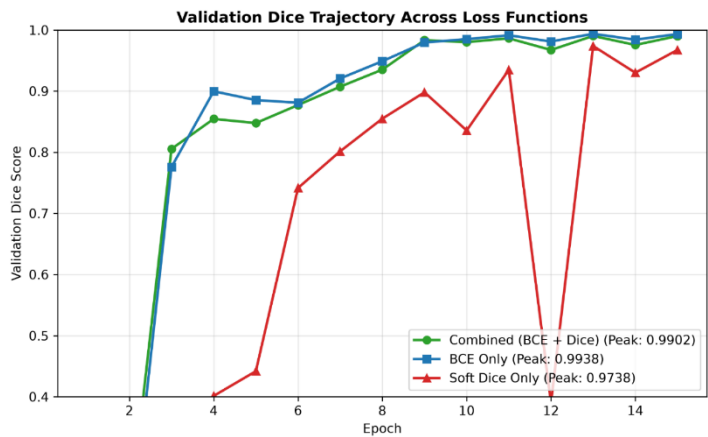


Figure 2: Validation Dice trajectory across the three loss functions. BCE achieved the highest peak Dice, while the combined loss also showed strong and generally stable convergence. Soft Dice was less stable, including a large reduction at epoch 12.

The results show that the combined loss was not numerically the best in this experiment. BCE-only achieved the highest validation Dice. The combined checkpoint was still kept in the hybrid pipeline because it explicitly includes both pixel-wise classification.

Q2. Did U-Net improve on classical Otsu segmentation for your modality?

Method	Mean Dice	Mean IoU
Classical Otsu	0.9776	0.9562
U-Net	0.9891	0.9784

Table 3: Mean segmentation performance of classical Otsu and U-Net across the 12 unseen test images. U-Net achieved higher mean Dice (0.9891 vs 0.9776) and IoU (0.9784 vs 0.9562), indicating better overall agreement with the ground-truth masks.

U-Net therefore provided better pixel-level segmentation across the test set.

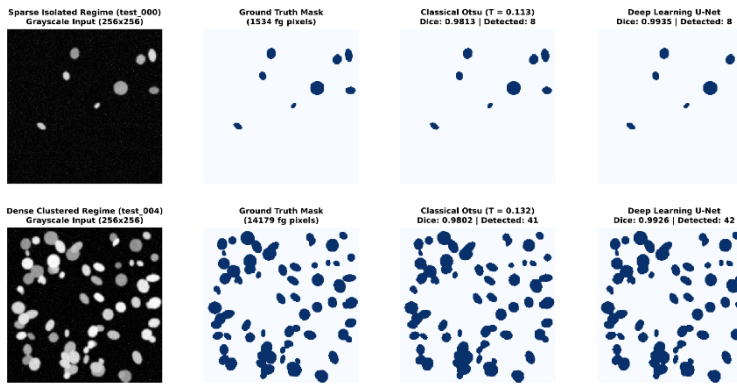


Figure 3: Comparison of ground truth, classical Otsu segmentation and U-Net predictions for sparse and difficult high-density examples

For the sparse image test_000, U-Net achieved Dice = 0.9935, compared with .9813 for Otsu. However, Otsu still detected the correct 8 connected objects, which shows that a simple classical method can still be useful for clear, well-separated objects.

For the high-density test_004, U-Net achieved Dice = .9926, while Otsu achieved about .9802. Otsu produced 41 connected components and U-Net produced 42. However, the ground-truth dataset contains 75 individual nuclei, which points to an important limitation: both methods can merge touching nuclei even when the binary segmentation mask is highly accurate.

Q3. Report your U-Net's Dice and IoU. What do these numbers mean, and where does the model tend to make its mistakes (Which images or regions)?

Across all 12 unseen test images, the final U-Net achieved:

Mean Dice = 0.9891 ± 0.0041

Mean IoU = 0.9784 ± 0.0080

Dice measures the overlap between predicted and ground-truth foreground regions, while IoU measures the intersection divided by the complete union. Both scores being close to 1 means very strong binary segmentation.

Still, these scores should not be seen as perfect nucleus counting. The test-set ground truth had an average of 33.4 nuclei, while the predicted masks produced an average of 20.8 connected components. The mean absolute counting error was 12.58 nuclei.

This difference is mainly due to touching nuclei being joined into one connected region. For example, test_010 contains 78 ground-truth nuclei, but the U-Net mask produces only 34 connected components, even though it achieves Dice 0.9866. This shows the difference between segmentation accuracy and instance-count accuracy.

The remaining pixel-level mistakes are mostly around object boundaries and areas where nuclei are close together. Therefore, an instance-separation step such as watershed or distance-transform post-processing would be needed for more reliable individual nucleus counting.

4. Hybrid pipeline, Hallucination Control and Clinical Trust

The final hybrid pipeline was tested on all 12 unseen test images. Every accepted image produced valid structured JSON, giving 100% schema validity. The LLM also reproduced the connected-component count provided by the deterministic analysis in 12/12 cases, giving a 100% LLM measurements-reproduction audit match.

This should not be mistaken for 100% true nucleus-count accuracy. The audit checks whether the LLM changed or invented the numerical count given to it. The separate ground-truth comparison shows that connected-component counting still underestimates nuclei in dense images.

Q4. Where in the pipeline can the LLM hallucinate, and what design choices reduce that risk? Why does keeping the structured JSON as the “Source of truth” “help?”

Hallucination can happen in two places. First, a direct VLM can misinterpret visual patterns or introduce unsupported biological statements. Second, a text LLM can add qualitative claims that are not justified by the measurements provided.

The final pipeline reduces these risks in several ways: the image is analyzed before the LLM, the prompt limits the model to supplied measurements, temperature is set to 0.0, structured fields are validated, model provenance is recorded, and numerical values are checked against the deterministic measurements.

Most importantly, the structured numerical record is treated as the source of truth. The narrative is only an interpretation of that record. This means key values such as connected-component count, area, and coverage can be checked independently instead of relying on generated prose.

Robustness experiment

Image	Condition	Intensity range	Coverage	components	Gate
test_000	Clean	0.3468	2.37%	8	Pass
test_000	Blur	0.1836	3.91%	7	Pass
test_000	Low contrast	0.0505	100%	1	Reject
test_004	Clean	0.3468	21.83%	42	Pass
test_004	Blur	0.2847	37.51%	15	Pass
test_004	Low contrast	0.0505	100%	1	Reject

Table 4: Robustness of the hybrid pipeline

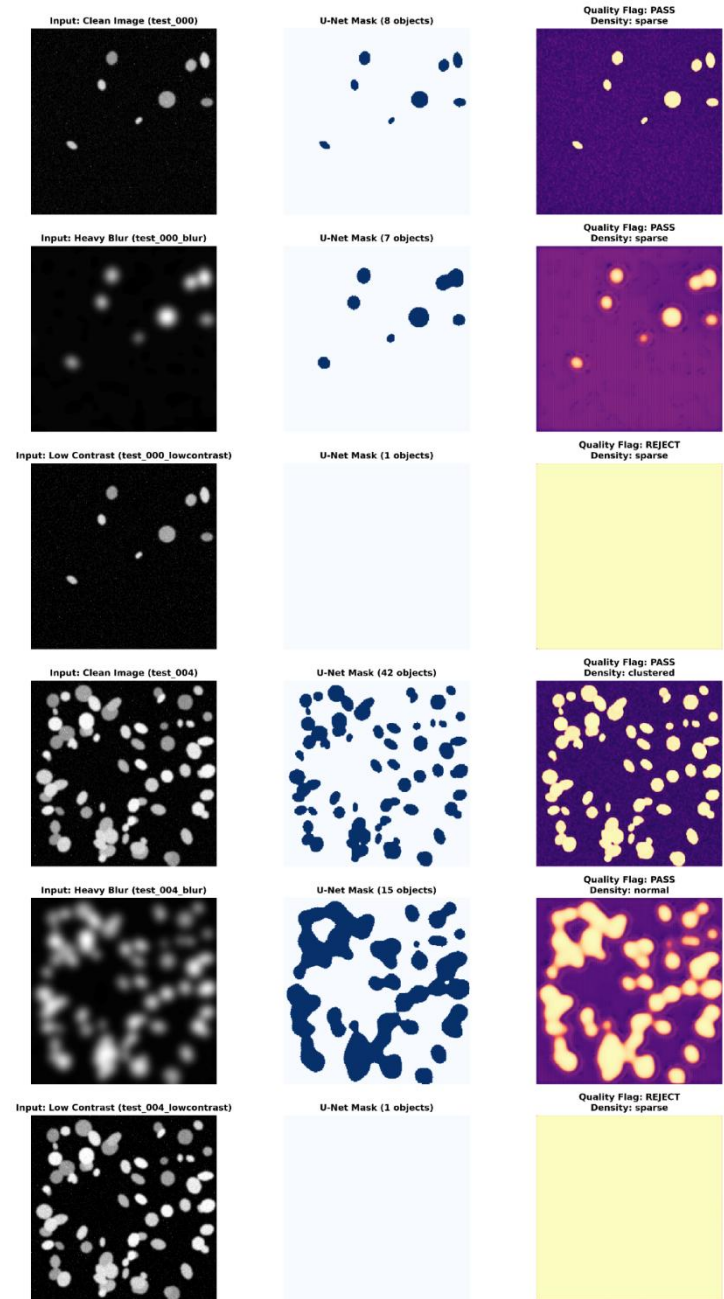


Figure 4: Robustness analysis showing clean, blurred and low-contrast inputs with their U-Net masks and prediction maps

Blur caused neighboring nuclei to merge, reducing test_004 from 42 to 15 connected components. The most severe failure happened under low contrast, where the model classified the whole image as one foreground region with 100% coverage.

The corrected pipeline now detects this before the LLM. If foreground coverage is above 75% or image contrast is below 0.8, the deterministic quality gate sets the results to reject and the LLM is not queried. This is important because it prevents

a clearly failed segmentation from being turned into a confident natural-language report.

Q5. Considering accuracy, auditability and the limits of the dataset, would you trust any part of this system in a real clinical setting? What single change would most improve trustworthiness?

I would not trust the current system for independent clinical decision-making. The U-Net has excellent binary segmentation performance on this dataset, and the structured pipeline is auditable, but the dataset contains only 112 synthetic images. Real microscopy images would have artefacts and biological variation.

I would trust the quantitative segmentation and morphometric stages only as a research or decision-support tool under expert supervision. I would not use the direct VLM or generated narrative as an independent diagnostic system.

The single most impactful improvement would be external validation using a much larger, real-world microscopy dataset with expert-verified annotations. This would show whether the strong Dice and IoU results generalize beyond the synthetic data. For nucleus counting specifically, instance-aware segmentation or watershed-based separation should also be added to handle merged touching nuclei.

5. Conclusion

The experiments show that the hybrid numbers-first design is more trustworthy than asking a VLM to interpret raw microscopy images directly. The U-Net achieved strong unseen-test segmentation performance (Dice 0.9891, IoU 0.9784) and outperformed Otsu at the pixel level. However, high segmentation overlap did not guarantee accurate individual nucleus counts in dense images.

The strongest part of the system is not the LLM alone, but the combination of deterministic measurement, structured JSON validation, provenance tracking, numerical audit checks, and pre-LLM quality gating. These controls make the results easier to inspect and lower the risk of unsupported generated conclusions.

6. References

- Ronneberger, O., Fischer, P. and Brox, T. (2015) ‘U-Net: Convolutional Networks for Biomedical Image Segmentation’, *MICCAI 2015*, pp. 234–241.
- Otsu, N. (1979) ‘A Threshold Selection Method from Gray-Level Histograms’, *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1), pp. 62–66.
- van der Walt, S. et al. (2014) ‘scikit-image: Image Processing in Python’, *PeerJ*, 2, e453.
- Caicedo, J.C. et al. (2019) ‘Nucleus segmentation across imaging experiments: the 2018 Data Science Bowl’, *Nature Methods*, 16, pp. 1247–1253.
- Rajpurkar, P. et al. (2022) ‘AI in health and medicine’, *Nature Medicine*, 28, pp. 31–38.